

Carney syndrome in a patient previously considered to have Peutz-Jeghers syndrome

To the Editor: We report a case of Carney syndrome in an 18-year-old male patient who presented with a previous diagnosis of Peutz-Jeghers syndrome and a rapidly growing asymptomatic tumor of his right upper lip, present for the past 7 months. Since age 6 years, this patient has had multiple lentigines of his face and oral mucosa. Approximately 3 years before coming to our attention, he had bilaterally enlarged testicles and a right orchiectomy that revealed a large cell calcifying Sertoli cell tumor. A review of systems also revealed occasional bloody stools. At that time, the patient was incorrectly diagnosed with Peutz-Jeghers syndrome and gastrointestinal surveillance was initiated. Endoscopy revealed a single 5-mm sessile polyp on the duodenal bulb that was not biopsied. Colonoscopy was unremarkable. Additional studies, including thyroid and endoscopic ultrasound as well as genetic evaluation, were not performed. His family history was negative for mucocutaneous lesions, gastrointestinal polyps, or myxomas.

When the patient presented at this institution, physical examination revealed a single, 7-cm, firm, nontender tumor involving the right upper lip with 2 flesh-colored, 0.5-cm papules overlying the surface. There were also multiple dark pigmented macules involving the face, lips, and oral mucosa. The tumor was biopsied and revealed a myxoid spindle cell proliferation with mild to moderate cellularity.

The patient underwent reconstructive surgery for complete excision. Gross examination revealed a 7- × 3.8- × 3.3-cm tumor. Microscopic examination demonstrated a hypercellular spindle cell proliferation with rich vascular network of thin-walled vessels, numerous neutrophils, and myxoid background that infiltrated adjacent connective tissue (Fig 1). Immunohistochemical staining was positive for vimentin and CD34 and negative for S100, AE1/AE3, factor XIIIa, and actin. The lesion was called a superficial angiomyxoma, the patient was correctly diagnosed with Carney syndrome, and proper follow-up was initiated.

This case highlights the importance of following strict diagnostic criteria as well as understanding the pathology behind the diagnostic criteria of Peutz-Jeghers and Carney syndrome (Table 1). The incorrect diagnosis of Peutz-Jeghers was made prematurely because diagnostic criteria had not yet been met.¹ This prompted inappropriate follow-up (gastrointestinal surveillance) looking for intestinal polyps. Peutz-Jeghers associated polyps are a disorganized

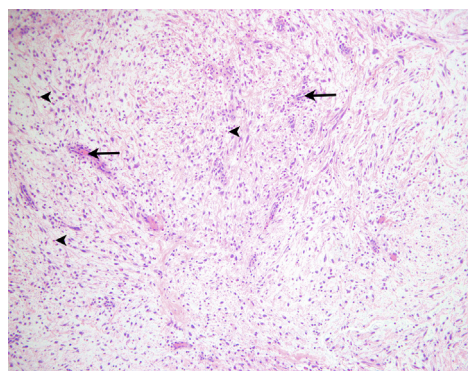


Fig 1. Carney syndrome; superficial angiomyxoma. Spindle-shaped cells within a copious, vascularized basophilic matrix (arrow) and myxoid stroma with numerous neutrophils (arrowhead).

growth of normal tissue (hamartomatous) with a smooth muscle core that is unique to Peutz-Jeghers polyps and not seen in patients with Carney syndrome.² Regardless of whether these polyps carry malignant potential, obtaining a biopsy helps fulfill diagnostic criteria when the diagnosis is still in question. Both syndromes share mucocutaneous lentigines as a feature but only Carney syndrome is associated with superficial angiomyxomas.³ It is important to recognize when these occur, because there is a higher risk of cardiac myxomas that can obstruct blood flow and cause heart failure.⁴ Physicians should rule out Carney syndrome and Peutz-Jeghers syndrome when a patient presents with features common to both. Finding one additional feature unique to either improves diagnostic certainty and guides the physician toward a proper follow-up and surveillance strategy. Recommended follow-up can help detect precursor lesions and other complications before they become life threatening.⁵

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Table I. Peutz-Jegher and Carney syndrome comparison chart. **Similar features in bold**

	Peutz-Jeghers	Carney
Diagnostic criteria	2 of 3 criteria: • Characteristic mucocutaneous pigmentation • Histologically confirmed PJS-type polyps • Family history of characteristic polyps or mucocutaneous pigmentation	1 criterion from clinical presentation and at least 1 additional criterion from either clinical presentation or tumors. (Seek 1 additional criterion if both criteria found are not unique to Carney.)
Genetics	Autosomal dominant , chr. 19p and 19q loci. Genes: <i>STK11 (LKB1)</i>	Autosomal dominant , chr. 2p16 and 17q22–24 loci. Genes: <i>PRKARIA</i> and others
Clinical presentation	• Mucocutaneous, pigmented spots of lips and buccal mucosa • Gynecomastia and growth acceleration	• Mucocutaneous pigmented spots of face, lips, eyelids, conjunctiva, and oral mucosa • Gynecomastia, growth acceleration, acromegaly
Tumors	• Sertoli cell tumors • Sex cord tumors with annular tubules • Various cancerous and precancerous lesions from: ○ Breast ○ Thyroid ○ Small intestine ○ Colon ○ Stomach ○ Pancreas ○ Esophagus ○ Ovary ○ Lung ○ Uterus ○ Uterine cervix	• Large cell calcifying Sertoli cell tumor • Breast myxomatosis • Breast ductal adenomas • Thyroid carcinoma • Myxomas: cutaneous, mucosal, and cardiac • Growth hormone producing adenoma • Psammomatous melanotic schwannomas • Blue nevus, epithelioid blue nevus • Osteochondromyxoma
Evaluation following initial diagnosis	• Genetics consultation • Men: testicular ultrasound • Women: gynecologic and breast examination • Upper endoscopy plus small bowel examination (MR or CT enterography, wireless capsule endoscopy) • Colonoscopy	• Genetics consultation • Men: testicular ultrasound • Women: transabdominal ultrasound • Imaging or biochemical screening for endocrine tumors • Thyroid ultrasound
Surveillance	• Annual history, physical, and routine blood work • Testicular exam (ultrasound if symptomatic) • Colonoscopy and upper GI endoscopy	• Annual history, physical, and routine blood work • Testicular exam (ultrasound if symptomatic) • Monitor linear growth rate and pubertal staging. • Annual or biannual echocardiograms • Further evaluation as indicated: ○ Diurnal cortisol levels ○ Dexamethasone stimulation test ○ 3-hour oral glucose tolerance test ○ 90-minute thyroid releasing hormone ○ MRI of the brain, spine, chest, abdomen, retroperitoneum, pelvis

Bold items indicate where Peutz-Jeghers and Carney syndrome are similar. When making a specific diagnosis, physicians should make sure they are using more than just the ones in bold to distinguish between the two syndromes.

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Focal nail dystrophy in a cutaneous nerve distribution after trauma

To the Editor: Dystrophic nail changes have been described in association with traumatic injury, and may be focal or generalized. The determinants of the pattern of nail dystrophy remain uncertain, although immobilization is thought to be a major contributor. We report a case wherein unilateral nail dystrophy appeared to follow a median nerve distribution after a thumb injury suggesting a neurogenic etiology for its development.

A 10-year old schoolgirl sustained a closed oblique fracture to the proximal phalanx of her left thumb (Fig 1) following a crush injury in a hinged gate. She had no other injuries and at no stage demonstrated any clinical evidence of nerve injury or direct nail or fingertip damage. Initial swelling and edema were confined to the thumb and subsided with elevation and immobilization of the thumb, hand, and wrist in a plaster cast for 2 weeks. Her fingers were excluded from the cast and were freely mobile.

She developed dystrophic nail changes (Fig 2) affecting the thumb, index, and middle fingers of the left hand; these were first apparent approximately 2 to 3 weeks after the original injury and progressed distally from the base of the nails. There were no other associated symptoms or signs. She regained a full range of movement in all digits. The appearance of the nails had resolved completely by approximately 6 months after the injury.

Nail changes have been described in the literature in association with a variety of local and systemic conditions, as well as following injury. Localized nail dystrophy has been described in association with spinal trauma¹ and with reflex sympathetic dystrophy²; the presence of Beau lines is often observed.



Fig 1. Initial radiologic appearance showing fractured proximal phalanx of left thumb.



Fig 2. Dystrophic nail changes to thumb, index, and middle fingers of left hand. (Photograph obtained approximately 18 weeks after the initial injury.)

There is evidence that nail plate growth may be accelerated by trauma and is slowed by immobilization.³ While localized nail changes similar to those reported here have been described after direct nerve injury,⁴ we have been unable to find any reports of nail growth disruption of this pattern without evidence of nerve damage. Anatomic studies suggest that innervation of the nail beds is highly variable, but in most cases the nail beds probably receive dual sensory input via the proper digital and dorsal cutaneous nerves from both the radial and either median or ulnar nerve.⁵ The pattern seen in our case and that of Ross and Ward⁴ suggests that the volar innervation may dominate.

Our case highlights the association of trauma with nail dystrophy, and the distribution supports a neurogenic influence rather than immobilization in its development. We suggest that neurotrophic factors released locally by the relevant cutaneous sensory nerve may be important here.

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